Security Metrics and Executive Reporting Report

Introduction

This practical exercise focused on evaluating Security Operations Center (SOC) performance using key security metrics and presenting the findings in an executive-friendly format. The objective was to calculate and analyze important indicators such as Mean Time to Detect (MTTD), Mean Time to Respond (MTTR), false positive rate, and dwell time.

The analysis was performed using Elastic Security for log visualization and dashboard design, while calculations and analysis were conducted using spreadsheet tools. The results were summarized in a structured executive report to support decision-making and improve SOC efficiency.

Tools Used

Tool	Purpose
Elastic Security	Metrics visualization and dashboard creation

Objective

The main objectives of this task were:

- To calculate SOC performance metrics (MTTD, MTTR, dwell time, false positives)
- To design a dashboard for monitoring security performance
- To analyze incident timelines and detection efficiency
- To present findings in an executive-level report

Metrics Dashboard Design

A conceptual dashboard was designed in Elastic Security to monitor SOC performance in real time.

Key Metrics Included

- Mean Time to Detect (MTTD) = **2 hours**

- Mean Time to Respond (MTTR) = **4 hours**
- False Positive Rate = **15%**

Dashboard Components

- Time-series visualization of log activity
- Incident detection and response timelines
- Alert classification overview
- False positive monitoring panel

Purpose

The dashboard enables analysts and management to:

- Quickly assess SOC efficiency
- Identify delays in detection and response
- Monitor alert quality and accuracy

Metrics Analysis

Dwell Time Calculation

Dwell time represents the duration an attacker remains undetected within a system.

- Initial Compromise Time: 0 hours
- Detection Time: 2 hours
- **Dwell Time: 2 hours**

Interpretation

- A dwell time of 2 hours indicates **moderate detection capability**
- Faster detection mechanisms could significantly reduce risk exposure
- Continuous monitoring and alert tuning are required to improve performance

Dwell Time Summary

The dwell time for the simulated incident was calculated as 2 hours, representing the duration between initial compromise and detection. This indicates moderate detection capability. Reducing dwell time through improved monitoring, automation, and alert correlation is essential to minimize attacker impact and enhance overall SOC performance.

Executive Report

The analysis of SOC performance metrics indicates that the Mean Time to Detect (MTTD) is currently 2 hours, while the Mean Time to Respond (MTTR) stands at 4 hours. These values suggest that incidents are detected within a reasonable timeframe, but response efficiency can be improved. A false positive rate of 15% highlights the need for better alert tuning to reduce unnecessary workload on analysts. The calculated dwell time of 2 hours indicates that threats remain undetected for a moderate duration, increasing potential risk exposure. To enhance SOC performance, it is recommended to implement advanced detection mechanisms, improve alert correlation, and adopt automation through SOAR solutions. Continuous monitoring and regular optimization of detection rules will further strengthen response capabilities and reduce overall security risks.

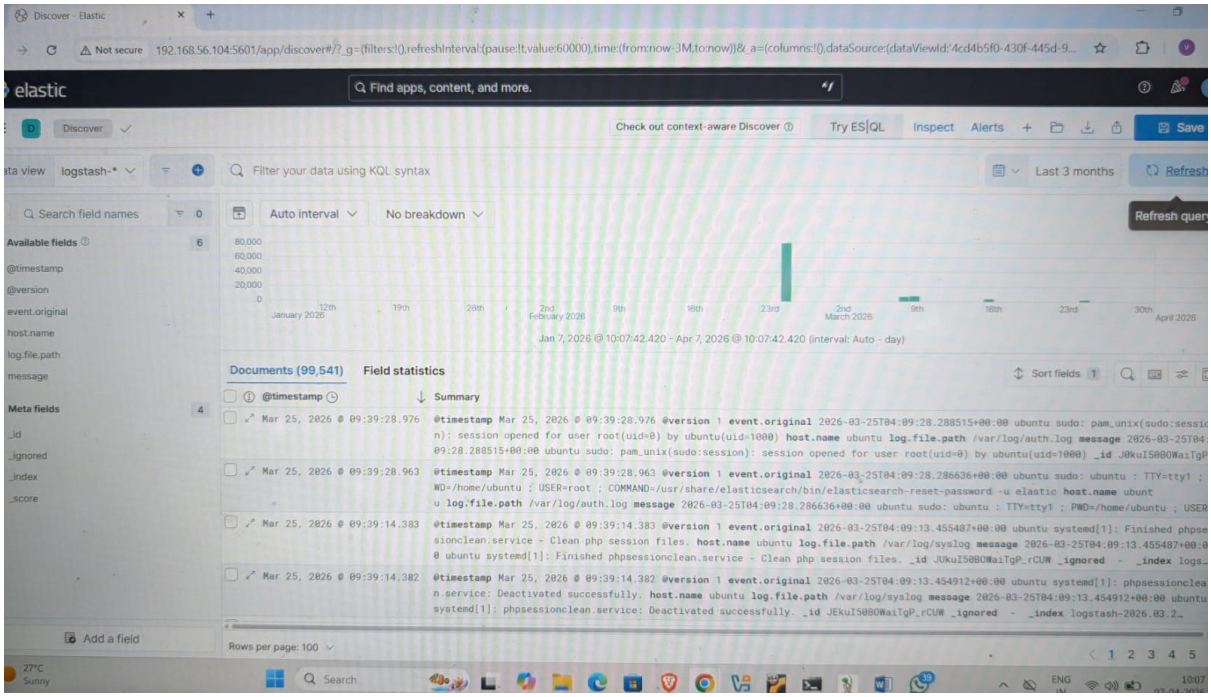
Recommendations

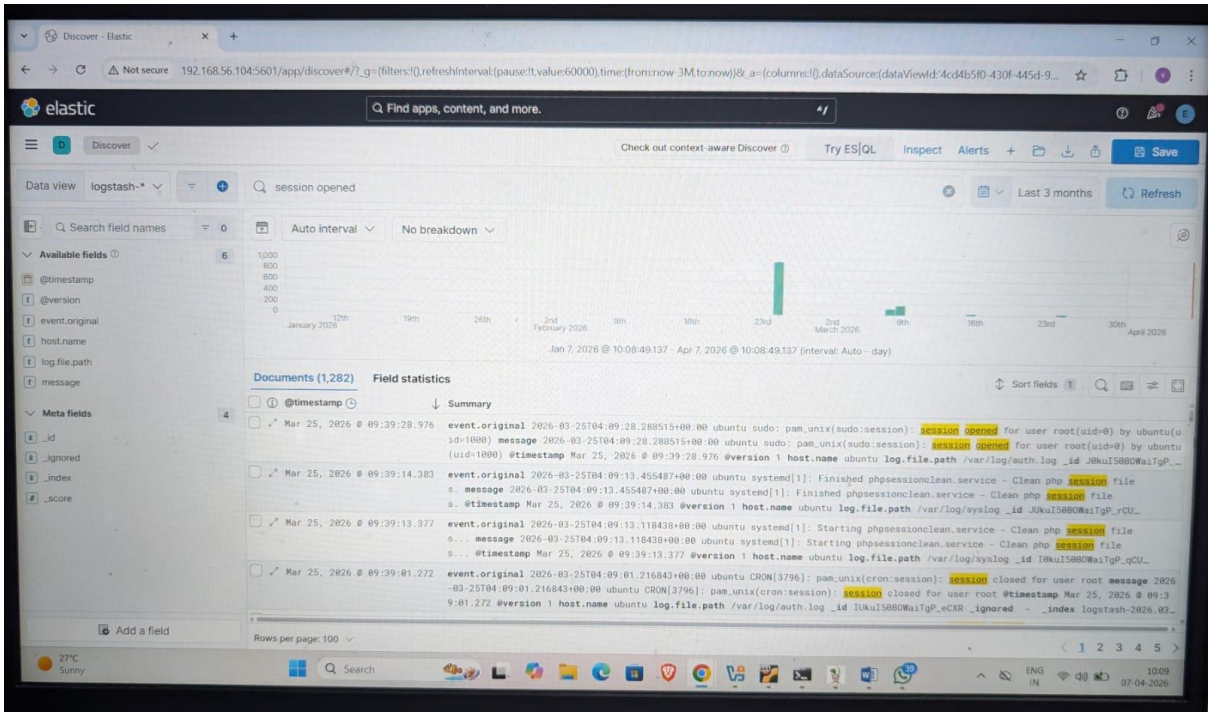
- Implement SOAR automation to reduce response time
- Fine-tune detection rules to lower false positives
- Enhance real-time monitoring and alert correlation
- Conduct regular SOC performance reviews
- Invest in advanced threat detection technologies

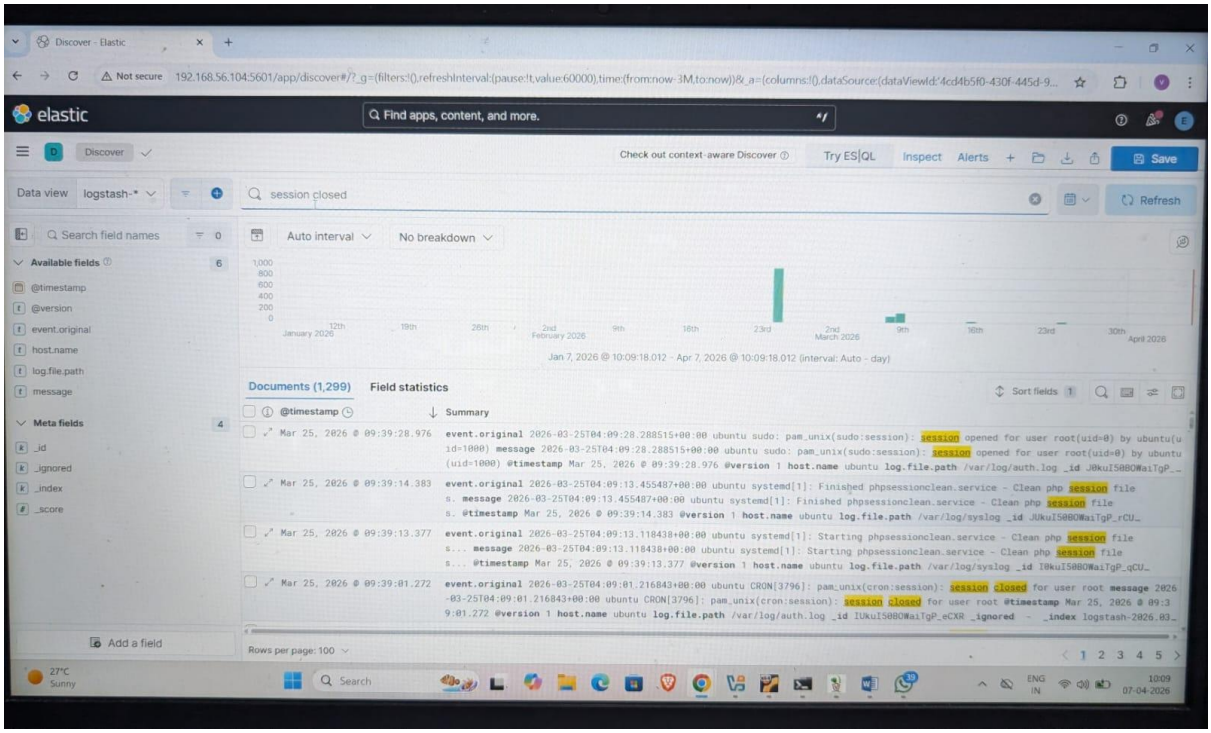
Conclusion

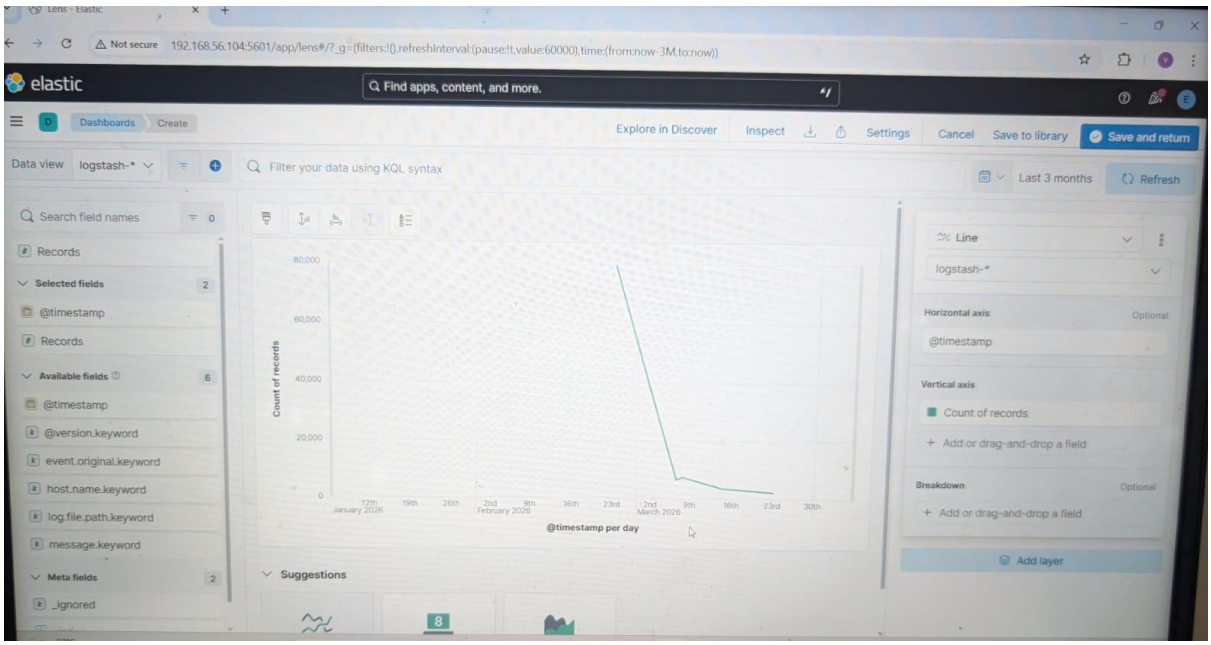
This practical exercise demonstrated the importance of security metrics in evaluating SOC performance. By analyzing key indicators such as MTTD, MTTR, false positive rate, and dwell time, it becomes possible to identify operational gaps and improve efficiency.

The use of Elastic Security for visualization and structured reporting enabled a clear understanding of SOC effectiveness. Executive reporting ensures that technical findings are translated into actionable insights, supporting better decision-making and continuous improvement in security operations.


Discover - Elastic
Not secure 192.168.56.104:5601/app/discover#/?_g=(filters:!(),refreshInterval:(pause:!t,value:60000),time:(from:now-3M,to:now))&_a=(columns:!(),dataSource:(dataViewId:'4cd4b5f0-430f-445d-9...
elastic
Find apps, content, and more.
Discover
Check out context-aware Discover
Try ES|QL
Inspect
Alerts
Save
logstash-*
Filter your data using KQL syntax
Last 3 months
Refresh
Search field names
Auto interval
No breakdown
Available fields
@timestamp
@version
event.original
host.name
log.file.path
message
Meta fields
_id
_ignored
_index
_score
Jan 7, 2026 @ 10:07:42.420 - Apr 7, 2026 @ 10:07:42.420 (interval: Auto - day)
Documents (99,541)
Field statistics
Sort fields
@timestamp
Summary
Mar 25, 2026 @ 09:39:28.976
Mar 25, 2026 @ 09:39:28.963
Mar 25, 2026 @ 09:39:14.383
Mar 25, 2026 @ 09:39:14.382
Add a field
Rows per page: 100


Discover - Elastic
Not secure 192.168.56.104:5601/app/discover#/?_g=(filters:!(),refreshInterval:(pause:!t,value:60000),time:(from:now-3M,to:now))&_a=(columns:!(),dataSource:(dataViewId:'4cd4b5f0-430f-445d-9...
elastic
Find apps, content, and more.
Discover
Check out context-aware Discover
Try ES|QL
Inspect
Alerts
Save
Data view
logstash-*
session opened
Last 3 months
Refresh
Search field names
Auto interval
No breakdown
Available fields
@timestamp
@version
event.original
host.name
log.file.path
message
Meta fields
_id
_ignored
_index
_score
Jan 7, 2026 @ 10:08:49.137 - Apr 7, 2026 @ 10:08:49.137 (interval: Auto - day)
Documents (1,282)
Field statistics
Sort fields
@timestamp
Summary
Mar 25, 2026 @ 09:39:28.976
Mar 25, 2026 @ 09:39:14.383
Mar 25, 2026 @ 09:39:13.377
Mar 25, 2026 @ 09:39:01.272
Add a field
Rows per page: 100


Discover - Elastic
Not secure 192.168.56.104:5601/app/discover#/?_g=(filters:!(),refreshInterval:(pause:!t,value:60000),time:(from:now-3M,to:now))&_a=(columns:!(),dataSource:(dataViewId:'4cd4b5f0-430f-445d-9...
elastic
Find apps, content, and more.
Discover
Check out context-aware Discover
Try ES|QL
Inspect
Alerts
Save
Data view
logstash-*
session closed
Refresh
Search field names
Auto interval
No breakdown
Available fields
@timestamp
@version
event.original
host.name
log.file.path
message
Meta fields
_id
_ignored
_index
_score
Jan 7, 2026 @ 10:09:18.012 - Apr 7, 2026 @ 10:09:18.012 (interval: Auto - day)
Documents (1,299)
Field statistics
Sort fields
@timestamp
Summary
Mar 25, 2026 @ 09:39:28.976
event.original 2026-03-25T04:09:28.288515+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000) message 2026-03-25T04:09:28.288515+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu (uid=1000) @timestamp Mar 25, 2026 @ 09:39:28.976 @version 1 host.name ubuntu log.file.path /var/log/auth.log _id J0kuI50BOWaiTgP_...
Mar 25, 2026 @ 09:39:14.383
event.original 2026-03-25T04:09:13.455487+00:00 ubuntu systemd[1]: Finished phpsessionclean.service - Clean php session file s. message 2026-03-25T04:09:13.455487+00:00 ubuntu systemd[1]: Finished phpsessionclean.service - Clean php session file s. @timestamp Mar 25, 2026 @ 09:39:14.383 @version 1 host.name ubuntu log.file.path /var/log/syslog _id JUkuI50BOWaiTgP_rCU...
Mar 25, 2026 @ 09:39:13.377
event.original 2026-03-25T04:09:13.118438+00:00 ubuntu systemd[1]: Starting phpsessionclean.service - Clean php session file s... message 2026-03-25T04:09:13.118438+00:00 ubuntu systemd[1]: Starting phpsessionclean.service - Clean php session file s... @timestamp Mar 25, 2026 @ 09:39:13.377 @version 1 host.name ubuntu log.file.path /var/log/syslog _id I0kuI50BOWaiTgP_qCU...
Mar 25, 2026 @ 09:39:01.272
event.original 2026-03-25T04:09:01.216843+00:00 ubuntu CRON[3796]: pam_unix(cron:session): session closed for user root message 2026-03-25T04:09:01.216843+00:00 ubuntu CRON[3796]: pam_unix(cron:session): session closed for user root @timestamp Mar 25, 2026 @ 09:39:01.272 @version 1 host.name ubuntu log.file.path /var/log/auth.log _id IUkuI50BOWaiTgP_eCXR _ignored - _index logstash-2026.03...
Add a field
Rows per page: 100
1 2 3 4 5
27°C Sunny
Search
ENG IN
10:09 07-04-2026


Lens - Elastic
Not secure 192.168.56.104:5601/app/lens#/?_g=(filters:!(),refreshInterval:(pause:!t,value:60000),time:(from:now-3M,to:now))
elastic
Find apps, content, and more.
Dashboards
Create
Explore in Discover
Inspect
Settings
Cancel
Save to library
Save and return
Data view
logstash-*
Filter your data using KQL syntax
Last 3 months
Refresh
Search field names
Records
Selected fields
@timestamp
Records
Available fields
@timestamp
@version.keyword
event.original.keyword
host.name.keyword
log.file.path.keyword
message.keyword
Meta fields
_ignored
80,000
60,000
40,000
20,000
0
Count of records
January 2026
February 2026
March 2026
@timestamp per day
Line
logstash-*
Horizontal axis
Optional
@timestamp
Vertical axis
Count of records
Add or drag-and-drop a field
Breakdown
Optional
Add or drag-and-drop a field
Add layer
Suggestions